(a) State Kirchhoff's first law.

....

.... [1]

(b) The intensity of light incident on Z is 130 W m^{-2}. The current in the battery is 38 mA.

(i) Show that the terminal potential difference of the battery is 4.8 V.

[2]

(ii) Calculate the current I_2 in Y.

I_2 = ... A [3]

(iii) Calculate the power dissipated in Y.

power = ... W [2]

(iv) The intensity of the light incident on Z decreases.

State and explain the effect on the terminal potential difference of the battery.

....

....

....

....

.... [3]

[Total: 11]

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